

DRY-RUN GUIDE | Bubble.py
Source: Dsa\sort\Bubble.py

CORE BEHAVIOR

Orders values using the comparisons, swaps, insertion, partition, or merges shown in this file.

REPRESENTATIVE INPUT

[4,1,3,2].

STEP TRACE

Each pass/partition places values nearer their final positions; completed sequence is nondecreasing.

EXPECTED OUTPUT / RETURN

1 2 3 4 (check source for exact formatting).

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Dsa\sort\Bubble.py; line numbers refer to the original file. No source was modified.

```
001 | def bubble_sort(arr):
002 |     n = len(arr)
003 |     for i in range(n):
004 |         swapped = False
005 |         for j in range(0, n - i - 1):
006 |             if arr[j] > arr[j + 1]:
007 |                 arr[j], arr[j + 1] = arr[j + 1], arr[j]
008 |                 swapped = True
009 |         if not swapped:
010 |             break
011 |     return arr
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.